

INSTITUTE OF ACTUARIES OF INDIA

Subject SP5 – Investment and Finance

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INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1: Company A Life uses derivatives to manage its reinvestment risk.**(i)** Reinvestment risk that A Life can manage using derivatives are:

1. Future contractual premiums that company is expected to receive - Life insurance companies receive continuous inflows of cash from policyholder premiums. To maintain profitability and meet future payouts (e.g., death benefits, annuities), these funds need to be invested in income-generating assets. If reinvestment options offer lower returns, the company may struggle to earn enough to cover future liabilities.
2. Reinvestment of maturity value of its bond holding - Life insurers typically invest in long-term bonds and fixed-income securities to match their long-term liabilities. If these investments mature during a period of declining interest rates, the company may have to reinvest the proceeds in lower-yielding assets, reducing the overall income from its investment portfolio.
3. Callable Bonds - If the company holds callable bonds (bonds that the issuer can repay before maturity), it faces the risk that issuers will call the bonds when interest rates decline. The company will then need to reinvest the returned principal at a lower interest rate.
4. Annuity Products - Many life insurance companies offer annuity products, which guarantee a certain rate of return to policyholders. If the general level of interest rates falls, the company may face difficulty generating sufficient investment returns to meet its guaranteed payout obligations, especially when reinvesting incoming premiums.

[4]

(ii) While derivatives can be an effective tool for managing and hedging risks, such as reinvestment risk, interest rate risk, and currency risk, they also expose life insurance companies to several risks. These risks arise from the complexity and nature of derivatives contracts and can have significant financial implications if not managed properly. Below are the key risks that a life insurance company may face when using derivatives for hedging:**1. Market Risk**

- **Risk:** It arises when the value of a derivative changes due to fluctuations in underlying asset prices (e.g., interest rates, currencies, or equity prices).
- **Impact:** If the hedging strategy is not perfectly aligned with the underlying exposure, market movements may lead to losses. This is particularly relevant if there is a **basis risk**, where the movement in the derivative's value does not perfectly offset the movement in the underlying exposure.

2. Counterparty Risk (Credit Risk)

- **Risk:** This is the risk that the counterparty to the derivative contract may default on its obligations, failing to make required payments or deliver the agreed-upon asset.
- **Impact:** If the counterparty defaults, the company may lose the hedge and be exposed to the original risk or may have to pay to replace the hedge at unfavorable terms, leading to financial losses.

3. Liquidity Risk

- **Risk:** Liquidity risk arises when a company is unable to exit a derivative position easily or quickly due to lack of market activity or unfavorable pricing.
- **Impact:** If the company needs to unwind or modify its derivative positions, it might face significant costs or be unable to do so in time to avoid losses. In times of market stress, derivative instruments may become illiquid, and the company might be forced to hold positions longer than anticipated.

4. Operational Risk

- **Risk:** Operational risk stems from the potential for failures in internal processes, systems, or human error in executing, monitoring, and managing derivative contracts.
- **Impact:** Poor execution, inadequate oversight, incorrect valuations, or mistakes in reporting derivative positions can lead to financial losses, regulatory penalties, or reputational damage.

5. Model Risk

- **Risk:** Derivatives often require complex pricing models, especially for more exotic instruments. Model risk arises if the models used to price derivatives or measure risk are incorrect or based on faulty assumptions.
- **Impact:** Incorrect valuations or inaccurate risk assessments can lead to unexpected losses. The use of flawed models might also lead to ineffective hedging strategies.

6. Basis Risk

- **Risk:** Basis risk arises when the movement in the value of the derivative does not exactly match the movement of the underlying exposure being hedged. This can occur when there is a mismatch between the characteristics of the derivative and the hedged item.
- **Impact:** Even though the company is hedging a particular risk, basis risk may result in partial or ineffective hedging, leading to financial losses.

7. Leverage Risk

- **Risk:** Derivatives often involve significant leverage, meaning that relatively small changes in the value of the underlying asset can result in large gains or losses on the derivative position.
- **Impact:** While leverage can amplify returns, it can also magnify losses. This can lead to substantial financial losses if market movements are unfavorable or if the company's exposure to leverage is not properly managed.

8. Collateral and Margin Risk

- **Risk:** Derivatives often require posting collateral or margin to cover potential future losses. Collateral risk arises when the company needs to provide additional cash or liquid assets to meet margin calls during periods of market volatility.
- **Impact:** If the market moves against the derivative position, the company may be required to post additional collateral, which can strain liquidity or force the sale of other assets at a loss.

9. Regulatory and Compliance Risk

- **Risk:** Life insurance companies are subject to strict regulatory requirements when using derivatives, including reporting, risk management practices, and capital requirements. Non-compliance with these regulations can result in penalties.
- **Impact:** Failure to meet regulatory standards could lead to fines, legal repercussions, and damage to the company's reputation. Derivative use might also increase capital requirements, affecting profitability.

10. Valuation Risk

- **Risk:** Many derivatives, particularly over-the-counter (OTC) derivatives, are not traded on exchanges and can be difficult to value accurately. Valuation risk arises when there is uncertainty or complexity in determining the fair market value of a derivative position.
- **Impact:** Inaccurate valuation can lead to incorrect financial reporting, misjudging the hedging effectiveness, or mismatches in risk management strategies.

While derivatives are powerful tools for hedging risks, they introduce their own set of risks, including counterparty, liquidity, operational, and valuation risks. Effective risk management practices, strong governance, robust modelling, and a clear understanding of the underlying exposures and risks are crucial to successfully using derivatives for hedging purposes.

(iii)

A. Key advantage of using FRAs are:

1. **Effective hedging against interest rate risk.**
2. **Customization** of terms to meet specific needs.
3. **No upfront costs**, making them cost-effective.
4. **Cash settlement** without physical delivery of assets.
5. They are highly **liquid instruments**
6. Improved **predictability of future cash flows**.
7. Help manage **reinvestment risk**.
8. Provide **liquidity** and ease of use in the OTC markets.
9. Allow hedging without the need to hold **underlying assets**.
10. Ideal for **short-term hedging** needs.

[4]

[17 Marks]

Solution 2: Areas that are covered in Factsheet which give insights into the fund for customers to understand the composition and risk along with the benchmark are the followings:

- 1) Fund objective
- 2) Inception date of the fund
- 3) NAV
- 4) AUM and split by key asset classes
- 5) Benchmark
- 6) Fund managers and no. of funds that they are managing
- 7) Asset classes – both fund objective and actual holding
- 8) Sector allocation
- 9) Returns – fund and benchmark
- 10) Return period – Since inception, 1,3,5,7,10 yrs (as applicable)
- 11) Key securities/bonds invested
- 12) Key sectors invested in
- 13) Rating profile of the fund, especially in debt funds
- 14) Risk parameters like yield, maturity, modified duration etc.
- 15) Commentary in case of any exception/key drivers of movement

[4 Marks]**Solution 3:**

- (i) A long-only, non-leveraged strategy is one where it is not possible to short sell (borrowed) assets (in anticipation of price falls) or to borrow to provide further capital for investment. This may be because the investor is prevented from doing so either by regulation or by the legal documentation governing the running of the investment fund.

[2]

(ii)

- 1) A hedge fund/absolute investment can be defined as an investment fund that aims to meet high or absolute returns by investing across a number of asset classes or financial instruments which will help the family office earn higher returns.
- 2) Hedge funds typically have less restrictions on thereby offering an avenue to take on more risk with the aim to get higher return
 - a. borrowing
 - b. short-selling
 - c. the use of derivatives
- 3) It greatly increases the opportunity set of available investment strategies, thereby offering opportunities for higher investment returns or provide diversification to the current portfolio, thereby enhancing the overall return of the portfolio
- 4) A hedge fund generates higher returned by

1. the placing of many aggressive positions on different assets
 2. a high level of borrowing given the limited size of the capital of the funds compared to the size of the individual investments
 3. a mix of investments for which the price movements would be expected mostly to cancel each other out, except for the positive effect the hedge fund is looking for
 4. a willingness to trade in derivatives, commodities and non-income bearing securities
 5. a higher risk tolerance than other funds.
- 5) Hedge funds can covers a wide spectrum of investment strategies although some of the more common ones are thereby giving exposure to diversified spectrum of risk.
1. global macro funds
 2. event-driven funds
 3. market-neutral funds
 4. multi-strategy funds.
- 6) Hedge funds often have complex legal structures and the fund structures are held offshore to minimise tax and regulatory requirements, this would be good for a family office
- 7) Hedge funds have historically been subject to less regulation and financial reporting which will help the family office although legislation is being introduced in a number of developed markets which will require far more onerous reporting in the future.
- [8]

(iii) Hedge funds can covers a wide spectrum of investment strategies and different hedge funds provide different benefits to the family office. Choice of hedge fund would depend upon

1. Risk Appetite of the family office
2. Tax status of the family office
3. Tax efficiency of the hedge fund
4. Liquidity need of the hedge fund
5. Investment objective and return expectation/requirement of the family office
6. Manager's experience and expertise in managing a hedge fund
7. Investment strategy of the hedge fund
8. Fees and Expenses structure of the hedge fund
9. Transparency and Reporting of the hedge fund
10. Performance and track record of the hedge fund
11. Need for diversification and Asset allocation of the family office
12. Alignment with Family office values

[3]

Key hedge fund and what they offer"

- a. Global macro funds - Global macro funds invest based on macroeconomic trends and geopolitical events, making them ideal for family offices that want exposure to global markets and opportunities. These funds can protect against inflation or deflation and take advantage of currency, commodity, and interest rate movements. These funds focus on big-picture trends, using a mix of assets such as currencies, commodities, bonds, and equities. Their ability to go long or short provides flexibility in both rising and falling markets.
- b. Event-driven funds - Event-driven strategies are attractive to family offices because they seek to exploit pricing inefficiencies that occur during corporate events, such as mergers, acquisitions, bankruptcies, and restructurings. This can generate strong returns in both rising and falling markets, offering non-correlated returns. Event-driven hedge funds focus on specific corporate events that can affect stock prices, often with low correlation to broader market movements. They

tend to be less volatile than other strategies and offer targeted opportunities for return.

- c. Market-neutral funds - Market neutral hedge funds aim to deliver steady, absolute returns irrespective of market conditions, making them ideal for family offices looking for low-volatility strategies. These funds typically seek to minimize market exposure while generating alpha through stock-picking or arbitrage. By maintaining balanced long and short positions in related securities, these funds aim to eliminate overall market risk, generating returns based on security selection rather than market direction.
- d. Multi-strategy funds. - Multi-strategy hedge funds offer diversification across various asset classes, strategies, and geographies, helping to spread risk and reduce exposure to any single asset or market. Family offices often prefer diversification to protect wealth and reduce volatility. These funds use different investment strategies, such as long/short equity, arbitrage, fixed income, and global macro, within a single fund. This provides flexibility and adaptability to different market conditions.

[4]

[7]

[17 Marks]

Solution 4:

(i)

1. Impact of retention will depend upon (i) current retention level and (ii) how much increase in retention is expected.
2. For NPar and Par products, impact of retention will also depend upon average SA of such policies. If the average SA of these policies are below the retention threshold, then impact of increasing retention will not have an impact on the amount to be retained. This will not result in any material impact on these liabilities and hence current ALM strategy should be fine.
 - 2.1. Even if the increase in retention does result in higher cash flows to be borne by the company, it still not have any material impact on the overall ALM strategy provided that earlier ALM strategy was covering entire liability profile and did not exclude reinsured liabilities.
 - 2.2. If current ALM excluded reinsured liabilities, then a higher retention would lead to ALM mismatches, extent of which will be determined after rerunning the liability projection assuming higher retention
3. Duration risk - Higher retention will increase the duration of the liability profile which could lead to duration mismatch. Extent of mismatch will depend upon the MV of assets and liabilities, duration of assets and risk appetite of the company
4. Interest rate risk - Interest rate risk will increase if more liabilities are retained. The impact will especially be seen on Protection portfolio where the impact of retention would be the maximum. Impact on NPAR and PAR is expected to be limited.
5. Liquidity Risk - Liquidity risk increase for the company and this will impact the proportion of cash/cash equivalent assets in the company. Impact will be assessed based on recalculated liquidity ratios.
6. Credit Risk - Impact on credit risk will depend upon the investment landscape of the country. If there are enough high quality assets available then, increase in cash flows (retained block) can be easily invested without any impact on the credit profile of the company, else there is a possibility that company may have to go down the credit profile to invest the additional liabilities.
7. Solvency Risk - Capital requirements may increase due to changes in portfolio. In the short term, there could be a capital requirement resulting from higher mismatch leading to solvency risk for the company.

8. Use of derivative and impact under adverse scenarios. -Due to change in liability profile which will result in reinvestment risk, to mitigate which use of derivative is likely to increase. Risk pertaining to derivative portfolio that may get more pronounced are:
 - 8.1. Availability of derivative
 - 8.2. Collateral management leading to liquidity risk
 - 8.3. Solvency risk resulting from adverse MTM movement under stress conditions
 - 8.4. P&L impact due to cost of entering derivative
9. How increased sales/profitability needs to be accounted for?
 - 9.1. What is the expected increase in profits and will the investment strategy for the same be aligned with current strategy for SH fund or if higher retention strategy is a tactical one then is a different short term ALM strategy is needed to manage this increased profit
10. Risk Appetite of the company in respect to mismatches, impact under stress scenario
 - Increase in the risk profile basis the points above clubbed with higher cost of managing the risk needs to be aligned with stated risk appetite of the company for interest and expense risk. If a change in risk appetite is required, then it may involve discussion with BOD
11. Impact on the valuation rate of interest due to change in portfolio yield, thereby negating any benefit of improved profitability. Increase will also impact the profit available for investment, thereby impacting the overall investment strategy of the company. Portfolio yield may come down if higher proportion is invested in cash/cash equivalent for due to portfolio rebalancing as required.
12. The impact of higher retention will not impact the ALM position of ULIP portfolio as ALM strategy of these portfolio is driven by their respective fund mandates.

[12]

(ii)

- I.** Mean-variance portfolio theory, sometimes called modern portfolio theory (MPT), specifies a method for an investor to construct a portfolio that gives the maximum return for a specified risk, or the minimum risk for a specified return.

The application of the mean-variance framework to portfolio selection falls conceptually into two parts:

1. First the definition of the properties of the portfolios available to the investor – the opportunity set.
Here we are looking at the risk and return of the possible portfolios available to the investor.
2. Second, the determination of how the investor chooses one out of all the feasible portfolios in the opportunity set.

This is the determination of the investor's optimal portfolio from those available.

The assumptions underlying portfolio theory are as follows:

1. All expected returns, variances and covariances of pairs of assets are known.
2. Investors make their decisions purely on the basis of expected return and variance.
3. Investors are non-satiated.
4. Investors are risk-averse.
5. There is a fixed single-step time period.
6. There are no taxes or transaction costs.
7. Assets may be held in any amounts, ie short-selling, infinitely divisible holdings, no maximum investment limits.

[5]

II. Some of the limitation of the portfolio theory are below:

1. By ignoring actuarial risk – the risk that the investor fails to meet his or her liabilities – the theory as presented here is severely limited.
2. According to MPT, variance of return and expected return are all that matter – this is a key assumption. Other key factors that might influence the investment decision in practice are ignored. These include:
 1. the suitability of the asset(s) for an investor's liabilities
 2. the marketability of the asset(s)
 3. higher moments of the distribution of returns such as skewness and kurtosis
 4. taxes and investment expenses
 5. restrictions imposed by legislation
 6. restrictions imposed by the fund's trustees.
3. It should be noted that the length of the time horizon, which is likely to vary between investors, is not specified in MPT.

[4]

[21 Marks]**Solution 5:****(i)** The formula for the investment index is:

$$I(t) = \frac{\sum_i N_{i,t} P_{i,t}}{B(t)}$$

where:

- $N_{i,t}$ is the number of shares issued for the i th constituent at time t
- $B(t)$ is the base value, or divisor, at time t
- $B(t)$ is obtained from $B(t - 1)$ through the chain-linking process.

[2]

(ii)**(a)** Using the above formula at $I(0)$, the value of $B(0)$ will come to 3.897. [2]**(b)** Using the above formulae at $I(1)$ and using the value of $B(0)$ as calculated above, the answer would be 1051.07 [2]**[6 Marks]****Solution 6:****(i)** maximization of net of tax returns will depend on:

1. the total rate of tax on an investment
2. how the tax is split between income and capital gains
3. when the tax is paid
4. whether the tax is deducted at source or has to be paid subsequently
5. the extent to which tax deducted at source can be reclaimed by the investor
6. the extent to which losses or gains can be offset between different investments or over different time periods for tax purposes.
7. What is the rate of tax on other asset classes which can be used as an alternative to buying residential property
8. Is there any prevailing losses which can be carried forward from previous years to help reduce the tax burden

[4]

(ii)

1. Anchoring is a term used to explain how people will produce estimates.

2. They start with an initial idea of the answer ('the anchor').
3. They then adjust away from this initial anchor to arrive at their final judgement.
4. Thus, people base perceptions on past experience or 'expert' opinion, which they amend to allow for evident differences to the current conditions.
5. The effects of anchoring are pervasive and robust and are extremely difficult to ignore, even when people are aware of the effect and aware that the anchor is ridiculous.

[3]

Other aspects which could be applicable in this case would be:

1. Framing - The way a choice is presented ('framed') and, particularly, the wording of a question in terms of gains and losses, can have an enormous impact on the answer given or the decision made. Changes in the way a question is framed of only a word or two can have a profound effect.
2. Estimating probabilities - mainly "Representative Heuristics" and "Availability"
3. Overconfidence - People tend to overestimate their own abilities, knowledge and skills. Mainly Hindsight bias or Confirmation bias
4. Effect of Options - Experimental evidence also suggests that the range of options or choices presented to people may influence their choices. Mainly "primary effect", "recency effect", Immediate option, regret aversion, status quo bias.

[5]

[8]

[12 Marks]

Solution 7:

(i) ESG stands for Environmental, Social, and Governance. It's a framework used to evaluate a company's commitment to and performance in areas beyond financial metrics. Here's a brief rundown of each component:

1. **Environmental:** This aspect focuses on how a company's operations impact the natural environment. It includes considerations like energy use, waste management, resource conservation, and overall environmental footprint. Companies might be evaluated on their efforts to reduce greenhouse gas emissions, manage natural resources responsibly, or address climate change.
2. **Social:** This component examines how a company manages relationships with its employees, suppliers, customers, and the communities where it operates. It includes factors like employee welfare, diversity and inclusion, labor practices, community engagement, and human rights.
3. **Governance:** This pertains to how a company is governed and managed. It covers aspects such as board structure, executive pay, business ethics, transparency, and shareholder rights. Effective governance aims to ensure that the company is run in a fair, ethical, and responsible manner.

Investors and stakeholders use ESG criteria to assess a company's long-term sustainability and ethical impact, believing that strong ESG performance can be indicative of lower risk and better financial performance over time.

[3]

(ii)

1. **Risk Management:** ESG factors can significantly impact a company's financial performance and long-term sustainability. Environmental risks (like climate change), social issues (such as labor practices), and governance problems (like corruption) can affect a company's profitability and stability. Investors are increasingly recognizing that incorporating ESG factors helps in identifying and mitigating these risks.
2. **Regulatory Pressure:** Governments and regulatory bodies worldwide are increasingly mandating greater disclosure of ESG-related information. For example, regulations in Europe and the U.S. are pushing companies to report on their ESG performance. Compliance with these regulations can affect investment decisions and strategies.

3. **Investor Demand:** There is growing interest among investors, particularly millennials and institutional investors, in aligning their investments with their values. Many investors want to support companies that contribute positively to society and the environment, not just those that offer financial returns.
4. **Performance and Returns:** Evidence is mounting that companies with strong ESG practices often perform better over the long term. Studies have shown that ESG investments can lead to higher returns and lower volatility, making them attractive to investors looking for sustainable performance.
5. **Reputation and Brand Value:** Companies that excel in ESG practices often enjoy a better reputation, which can translate into customer loyalty, enhanced brand value, and competitive advantage. Investors recognize that strong ESG performance can enhance a company's reputation and market position.
6. **Long-Term Focus:** ESG investing aligns with a long-term investment horizon. By focusing on sustainability and ethical practices, investors are looking to support companies that will thrive over the long term, rather than those that might offer short-term gains at the expense of future sustainability.
7. **Stakeholder Pressure:** Companies are facing increasing pressure from various stakeholders—including customers, employees, and activists—to adopt sustainable and ethical practices. This pressure can drive companies to improve their ESG performance, making it a critical factor for investors to consider.

[4]

- (iii) Launching an ESG ULIP (Unit Linked Insurance Plan) fund involves several critical factors to ensure that the fund aligns with ESG principles while also meeting regulatory requirements and investor expectations. Here's a comprehensive list of factors to consider:

ESG Criteria and Framework

1. **Defining ESG Criteria:** Establish clear and transparent ESG criteria for selecting companies or assets. This could involve specific benchmarks for environmental, social, and governance performance (e.g., low carbon emissions, diversity in leadership, corporate governance standards).
2. **Data Sources:** Ensure access to reliable ESG data from third-party providers or proprietary research to evaluate companies' ESG performance. It is important to select source which is credible and acceptable to investment domain in general
3. **Materiality Assessment:** Prioritize ESG factors based on their relevance and potential impact on financial performance and long-term sustainability in specific sectors or industries.

Investment Strategy:

4. **ESG Integration:** Decide how ESG factors will be integrated into the fund's investment strategy. Will it exclude companies with poor ESG performance (negative screening), or include companies actively improving their ESG practices (positive screening)?
5. **Thematic Investing:** Consider focusing on specific ESG themes such as clean energy, sustainable agriculture, or social equity.
6. **Active vs. Passive Management:** Choose whether the ESG ULIP will be actively managed (with regular changes based on ESG assessments) or passively managed (tracking an ESG index).
7. **Long-term Focus:** Align the fund with long-term investment horizons, as ESG issues often unfold over longer periods.

Risk Management and Compliance

8. **Regulatory Requirements:** Comply with regulatory guidelines set by financial authorities on ESG disclosures, transparency, and reporting (e.g., SEBI's Business Responsibility and Sustainability Reporting in India, SFDR in Europe).
9. **Mitigating ESG Risks:** Ensure that the fund considers risks related to climate change, regulatory changes, social unrest, or governance scandals that could affect portfolio performance.
10. **Transparency and Reporting:** Establish a robust framework for ESG reporting, offering regular updates to policyholders about the ESG performance of the fund's portfolio.

Fund Structure and Product Design

11. **Fund Objective:** Clearly define the fund's objective, focusing on both financial returns and ESG impact. Communicate this to potential customers/policyholders effectively.
12. **ULIP Structure:** As a ULIP, the fund combines life insurance and investment. Ensure the structure allows for competitive premium allocation, mortality benefits, and policyholder flexibility (e.g., switching between ESG and non-ESG funds).
13. **Cost Structure:** Keep expense ratios competitive while factoring in costs related to ESG research, active management, and compliance.

ESG Benchmark or Index

14. **Benchmark Selection:** Select an ESG benchmark or index to track the fund's performance. This could be a widely recognized index like the MSCI ESG Leaders Index or a customized benchmark specific to the fund's ESG focus.
15. **Customization of Benchmarks:** Consider creating custom ESG indices based on regional or thematic needs, to better align with the fund's objectives.

Stakeholder Engagement

16. **Investor Education:** Educate potential customer about the importance of ESG investing and its potential long-term benefits, both in terms of returns and societal impact.
17. **Policyholder Communication:** Provide clear and accessible information on how the fund contributes to ESG objectives, helping policyholders understand the social and environmental impact of their investments.
18. **Corporate Engagement:** Actively engage with companies in the fund's portfolio to encourage better ESG practices, often called shareholder activism or stewardship.

Marketing and Distribution Strategy

19. **Sales Impact:** Identify and target customer who are inclined towards responsible investing or want to align their financial goals with their values. ESG ULIP funds may appeal to younger, more socially conscious investors or institutional investors with an ESG mandate. Discussion with sales team will help assess the potential topline that such a fund can get.
20. **Digital Platforms:** Utilize digital channels to market the ESG ULIP fund, providing transparency and ease of access to ESG reports, fund performance, and investment processes.
21. **Third-Party Ratings:** Consider acquiring ESG certifications or ratings from trusted third parties (e.g., Morningstar, Sustainalytics) to bolster the fund's credibility.

Performance Measurement

22. **Financial and ESG Returns:** Measure performance not only in terms of financial returns but also in the achievement of ESG goals. Establish KPIs for both financial and ESG outcomes.
23. **Impact Measurement:** Consider using impact measurement frameworks to assess how the fund has contributed to positive environmental or social outcomes, such as CO2 emissions reduced, gender diversity in leadership, etc.

24. **Reporting on Impact:** Provide transparent and regular reports to investors on both financial performance and ESG impact to build trust and credibility.

Lifecycle Management and Flexibility

25. **Fund Flexibility:** Ensure policyholders have options to switch between ESG and other types of funds within the ULIP structure, based on their evolving investment preferences and market conditions.
26. **Adapting to Changing ESG Trends:** ESG trends and standards evolve over time. Make provisions for updating the fund's strategy and criteria based on emerging risks, opportunities, and regulatory frameworks (e.g., new climate regulations or social norms).

Partnerships and Collaboration

27. **Collaborating with ESG Experts:** Consider partnering with ESG experts, consultants, or academic institutions to ensure that the fund adheres to best practices in ESG investing.
28. **Third-Party ESG Providers:** Work with specialized ESG data providers and rating agencies to ensure accurate and comprehensive ESG assessments.

Above would be important factors that one should cover in their report on creating an ESG ULIP fund that appeals to responsible customers, aligns with global sustainability goals, provides long-term value to policyholders and is a potential product that helps company achieve high new business growth.

[16]

[23 Marks]
